Promoters and catalyst poison

In a catalysed reaction the presence of a certain substance increases the activity of a catalyst. Such a substance is called a promoter.

For example in the Haber's process of manufacture of ammonia, the activity of the iron catalyst is increased by the presence of molybdenum. Hence molybdenum is called a promoter. In the same way Al_2O_3 can also be used as a promoter to increase the activity of the iron catalyst.

On the other hand, certain substances when added to a catalysed reaction decreases or completely destroys the activity of catalyst and they are often known as catalytic poisons.

Few examples,

In the reaction, $2SO_2+O_2 \rightarrow 2SO_3$ with a Pt catalyst, the poison is As_2O_3

i.e., As_2O_3 destroys the activity of Pt. As_2O_3 blocks the activity of the catalyst. So, the activity is lost.

In the Haber's process of the manufacture of ammonia, the Fe catalyst is poisoned by the presence of H_2S.

In the reaction, $2H_2+O_2 \rightarrow 2H_2O$,

CO acts as a catalytic poison for Pt - catalyst

Auto catalysis

In certain reactions one of the products formed acts as a catalyst to the reaction. Initially the rate of reaction will be very slow but with the increase in time the rate of reaction increases.

Auto catalysis is observed in the following reactions.

$$CH_3COOC_2H_5+H_2O \rightarrow CH_3COOH+C_2H_5OH$$

Acetic acid acts as the autocatalyst

$$2AsH_3 \rightarrow 2As+3H_2$$

Arsenic acts as an autocatalyst

Negative Catalysis

In certain reactions, presence of certain substances, decreases the rate of the reaction. Ethanol is a negative catalyst for the following reaction.

(i) $4CHCl_3+3O_2 \rightarrow 4COCl_2+2H_2O+2Cl_2$

Ethanol decreases the rate of the reaction

(ii) $2H_2O_2 \rightarrow 2H_2O+O_2$

In the decomposition of hydrogen peroxide, dilute acid or glycerol acts as a negative catalyst.

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